

#### **Element F: Consideration of Design Viability**

The design is realistic and nearly met all task requirements. The only failure—buoyancy imbalance from tether drag—was predictable based on missed force calculations. With polystyrene blocks and a floated tether, this flaw will be eliminated.

We used past MATE designs and Blue Robotics test logs to confirm viability. A risk matrix was created ranking: thruster failure, tether drag, unbalanced load, and controller lag. Mitigation steps are now planned in the next cycle.